

1 Market-wide liquidity-shock case study

1.1 Objective and hypotheses

The purpose of the case study is to test whether a liquidity shock applied to a small set of books affects otherwise unshocked books through a common inventory-capacity constraint. It is distinct from the scaling study. The scaling study asks whether one large coupled market can be executed efficiently; the case study asks whether the implemented coupling generates an economically interpretable cross-asset response.

Four predeclared hypotheses guide the analysis. First, aggressive sell orders should deteriorate liquidity in the targeted books. Second, if a globally constrained shared market maker is active, non-target books should exhibit a larger deterioration than under an uncoupled market maker. Third, this difference should be stronger under the tighter global capacity. Fourth, clusters with wider spreads and lower depth should exhibit a larger response than more liquid clusters. The first hypothesis concerns successful shock injection; the remaining hypotheses concern cross-book transmission.

1.2 Experimental design

The experiment uses the 30 January 2020 development-validation configuration for the balanced panel of 1,480 NASDAQ symbols. Each symbol is represented by one limit order book and retains its symbol-specific background-order-flow inputs. Behavioural policies are frozen at the level of ten liquidity clusters. The 6.5-hour regular session contains 23,400 one-second decision windows. Fifteen symbols are selected with a fixed mask that covers every cluster. At $t^* = 11,700$ seconds, an aggressive sell order equal to one contemporaneous top-of-book depth is submitted to each selected book.

The treatments are: (i) a globally constrained shared market maker with capacity 25 per asset; (ii) the same mechanism with capacity 100; (iii) an uncoupled specification that retains the liquidity-supply logic but removes the common cross-asset capacity constraint; and (iv) a control without the shared market maker. Every treatment is run with shock on and shock off for five entity-keyed seeds. Shock and control paths use common random numbers, giving 40 full-session paths in total. Each path uses 32 MPI ranks. The target mask is fixed across treatments, but the number of submitted shares is state dependent because it is defined as a multiple of contemporaneous depth. Consequently, differences in the direct response across treatments are descriptive and are not interpreted as responses to an identical share order.

For outcome Y , let $Y_{mcs}^1(t)$ and $Y_{mcs}^0(t)$ denote the mean of non-target books in mode m , cluster c and seed s in shocked and control paths. Spread deterioration is

$$\Delta_{mcs}^S(t) = S_{mcs}^1(t) - S_{mcs}^0(t), \quad (1)$$

whereas depth deterioration is

$$\Delta_{mcs}^D(t) = D_{mcs}^0(t) - D_{mcs}^1(t). \quad (2)$$

Positive values therefore indicate worse liquidity. The cross-asset estimand is the paired cross-mode contrast, where g and u denote the global and uncoupled modes,

$$\tau_{cs}^Y(t) = \Delta_{g,cs}^Y(t) - \Delta_{u,cs}^Y(t), \quad (3)$$

summarised over the 1,800 seconds following the shock. This construction removes seed-specific background variation and isolates effects attributable to the global coupling.

1.3 Liquidity-cluster composition

Table 1 reports unshocked simulated baselines. The cluster numbers are identifiers, not an ordinal liquidity ranking. The descriptive labels use spread and displayed depth only; the clustering itself also uses activity, return variance and price information. Spread is reconstructed from the asset summaries as $10^6 \bar{s}_a / p_a$ basis points because one exchange tick corresponds to 100 units of the ITCH 10^{-4} -dollar price grid. This conversion reconciles the asset summaries with the market-wide per-second spread series.

Table 1: Liquidity-cluster profiles and fixed shock targets. Baselines are means of five uncoupled, unshocked paths.

Cluster	Books	Descriptive profile	Spread (bps)	Top depth	Target symbols
0	146	widest, shallow	110.8	396	NBN
1	144	wide, medium depth	69.5	912	ELOX
2	191	moderate, shallowest	41.0	346	AVAV, ZBRA
3	266	moderate, shallow	36.1	527	BWB, LFVN, SPAR
4	161	wide, deep	73.9	2561	ITI, PXLW
5	196	tight, moderate depth	22.1	616	PRAA, PRFT
6	145	tight, deep	21.8	1504	OXLC
7	82	tightest, shallow	7.1	409	FTNT
8	79	moderate, exceptionally deep	29.1	11574	RFIL
9	70	tightest, deep	7.7	1987	FOX

The panel is deliberately heterogeneous. Cluster 0 has the widest spread, whereas clusters 7 and 9 have the tightest spreads. Cluster 8 has an order of magnitude more displayed depth than most clusters. The target design includes one to three symbols per cluster, so it provides broad coverage but not enough independent target books to estimate a precise cluster-specific direct response.

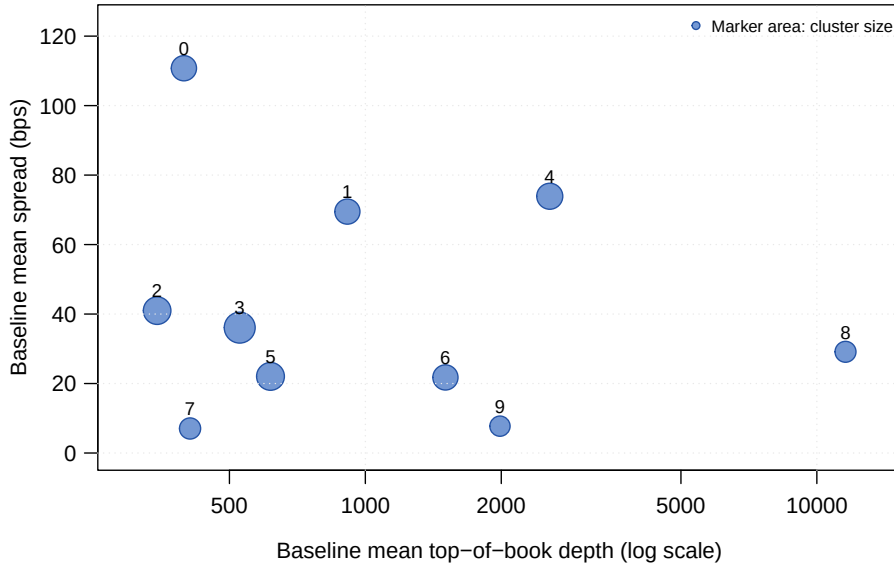


Figure 1: Unshocked cluster profiles. Marker area represents the number of books; labels identify clusters.

1.4 Shock execution and direct response

All requested shock quantities executed in every treatment. Table 2 reports mean targeted-book spread effects, with sample standard deviations across the five seeds in parentheses. The one-

second column is the immediate paired response; the 60- and 300-second columns are averages over $(0, h]$, not point estimates at the endpoint.

Table 2: Shock execution and paired targeted-book spread response.

Mode	Capacity	Mean total requested	Executed (%)	Shared absorption (%)	1 s	First 60 s	First 300 s
Global	25	10951	100	0.0	11.796 (0.541)	5.770 (1.516)	2.064 (1.748)
Global	100	8955	100	0.0	12.345 (1.935)	8.020 (2.091)	3.413 (0.378)
Absent	100	10459	100	0.0	12.422 (2.635)	5.492 (1.185)	2.655 (1.341)
Uncoupled	100	9504	100	9.7	0.062 (0.138)	0.243 (0.309)	0.353 (0.792)

The shock therefore enters the simulated market. Under global capacity 25, the targeted spread rises by 11.80 basis points after one second and by an average of 2.06 basis points over the first five minutes. The corresponding values under global capacity 100 are 12.35 and 3.41 basis points. Depth effects are considerably noisier and sometimes negative because background and local liquidity replenishment can add displayed quantity after the aggressive order. The uncoupled market maker absorbs 9.69% of executed shock volume on average; the globally constrained maker absorbs none.

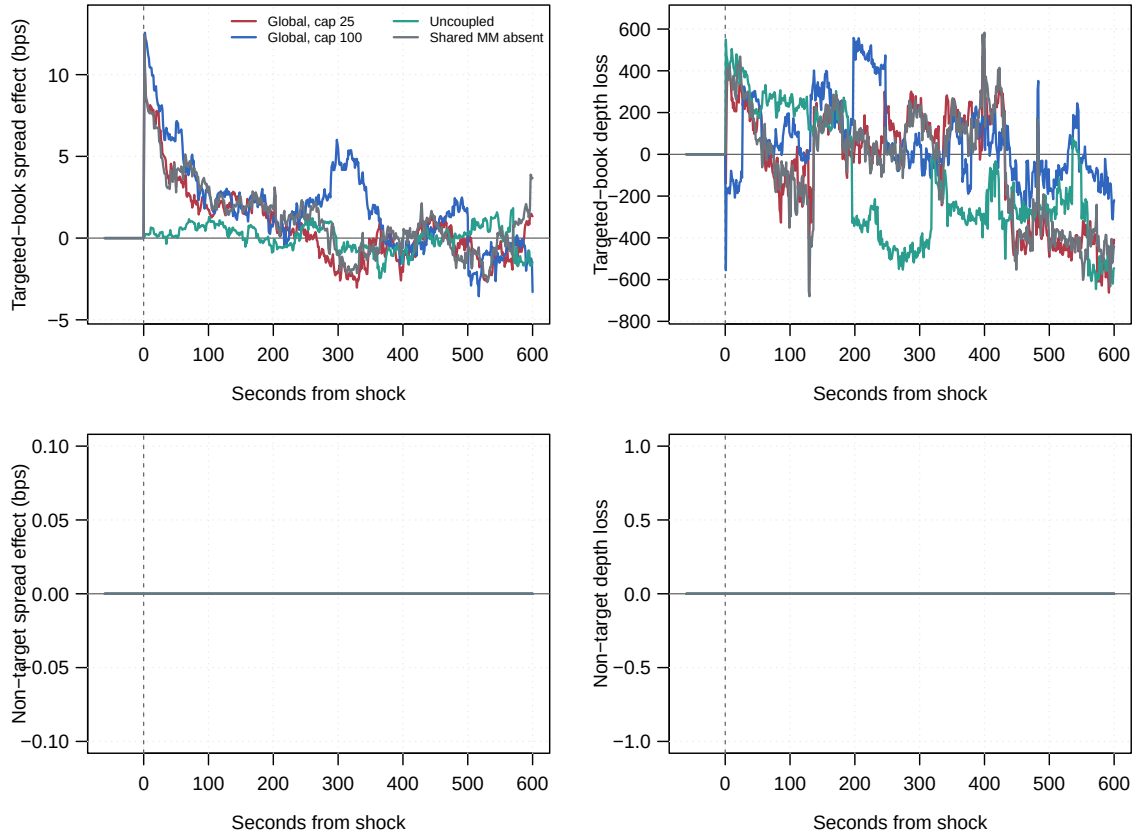


Figure 2: Paired shock-minus-control response averaged across five seeds. Non-target effects are exactly zero in every recorded window.

1.5 Cluster-level cross-book response

Table 3 gives the non-target response separately for every cluster and both global capacities. Each value is zero. This is not a rounded result: shock and control observations are bit-for-bit identical for all non-target books at every recorded second, seed and treatment. The same zero is obtained for full-session asset summaries and for the 30-minute post-shock endpoint. There is consequently no estimable difference between wide-spread, shallow clusters and tight-spread

or deep clusters. Hypothesis 4 is not supported, but the reason must be established before the zero can be given an economic interpretation.

Table 3: Mean paired non-target liquidity deterioration by cluster.

Cluster	Non-target books	Capacity 25		Capacity 100	
		Spread (bps)	Depth loss	Spread (bps)	Depth loss
0	145	0.000	0.0	0.000	0.0
1	143	0.000	0.0	0.000	0.0
2	189	0.000	0.0	0.000	0.0
3	263	0.000	0.0	0.000	0.0
4	159	0.000	0.0	0.000	0.0
5	194	0.000	0.0	0.000	0.0
6	144	0.000	0.0	0.000	0.0
7	81	0.000	0.0	0.000	0.0
8	78	0.000	0.0	0.000	0.0
9	69	0.000	0.0	0.000	0.0

1.6 Mechanism audit

The state of the shared market maker explains the cluster-level null. As shown in Table 4, the global quote scale becomes permanently zero long before t^* . At capacity 25 this occurs after 2.41 minutes on average; at capacity 100 it occurs after 23.62 minutes. The shock is applied after 195 minutes. The shared maker therefore supplies no liquidity and absorbs no shock volume when the intervention occurs.

Table 4: Pre-shock audit of the globally constrained shared market maker.

Capacity	First reduction (s)	Permanent zero (min)	Pre-shock zero (%)	Scale at shock	Absorption (%)
25	8.0	2.41	98.8	0.0	0.0
100	51.2	23.62	87.9	0.0	0.0

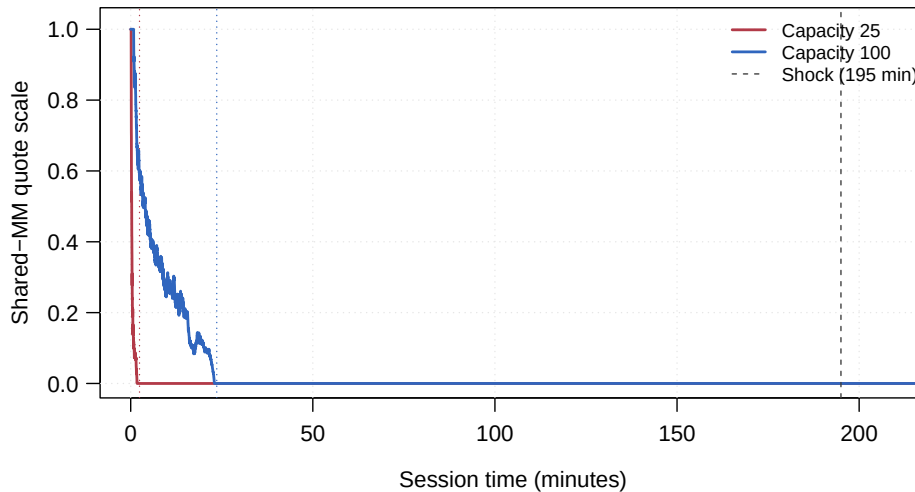


Figure 3: Median shared-market-maker quote scale. Both globally constrained treatments enter the zero-quote state well before the shock.

The implementation multiplies both risk-increasing and risk-reducing quotes by the global scale. Once that scale reaches zero, the dealer cannot trade to reduce inventory; its inventory

and utilisation cease to recover, making the zero-quote state absorbing. The primary difference-in-differences estimate is therefore non-identifying for shared-liquidity contagion. It shows that the implemented channel was inactive, not that an active common dealer prevented or failed to transmit the shock. Hypotheses 2–4 cannot be evaluated from this matrix.

1.7 Computational evidence

The rank-equivalence preflight processed 3,100,370 orders and produced the same state hash on one and 32 ranks. Runtime fell from 8.412 to 0.638 seconds, a speedup of 13.19 and efficiency of 41.2%. This establishes that whole-book decomposition changes execution time without changing the simulated state. The 40 production paths processed 6,030,761,053 orders in total.

The science matrix is not used as the principal scaling benchmark. Fourteen paths experienced intermittent communication stalls of 235–471 seconds while their compute time remained comparable with other paths. Treatment wall times would therefore confound the financial design with system noise. Strong- and weak-scaling conclusions are taken from the dedicated benchmark campaign, whereas this section uses MPI to make the large coupled market practical and to preserve causal equivalence across ranks.

1.8 Limitations and case-study conclusion

Three limitations determine the interpretation. First, the 2020 session is a development validation date rather than an untouched final holdout. Second, five seeds and one to three targets per cluster are insufficient for a precise cluster-specific direct-effect ranking. Third, the shared dealer is inactive at the shock, so the central contagion mechanism is absent. The cluster audit is nevertheless informative: it demonstrates that the null is universal across heterogeneous books and identifies the model state responsible for it.

The experiment therefore establishes successful shock injection, deterministic MPI equivalence and broad cluster coverage, but it does not establish cross-asset liquidity contagion. A corrected dealer should continue to submit risk-reducing inventory-unwind quotes at the global boundary while suppressing only risk-increasing quotes. A new pilot should require a positive quote scale immediately before the intervention and positive shared-maker absorption before the production matrix is repeated.

.1 Descriptive targeted-book effects by cluster

Table 5 reports full-session shock-minus-control spread differences for the targeted books. These values are supplementary diagnostics, not cluster-specific causal estimates. They average the intervention over the complete 23,400-second session, each cluster contains only one to three target symbols, and submitted quantities vary with contemporaneous depth. The per-second output aggregates all 15 targets and therefore cannot recover cluster-specific response trajectories after the run.

Table 5: Descriptive full-session targeted-book spread effects by cluster. Values are means with sample standard deviations in parentheses.

Cluster	Target symbols	Global 25	Global 100	Uncoupled	Absent
0	NBN	0.457 (4.842)	-6.011 (10.049)	0.184 (0.411)	1.898 (3.026)
1	ELOX	2.795 (8.368)	-2.270 (2.683)	0.342 (2.436)	3.225 (8.133)
2	AVAV, ZBRA	-0.325 (0.775)	0.247 (0.851)	-0.171 (0.312)	-0.460 (0.699)
3	BWB, LFVN, SPAR	0.080 (0.933)	-0.098 (0.435)	0.073 (0.296)	0.185 (0.780)
4	ITI, PXLW	-0.174 (0.662)	0.160 (0.696)	-0.005 (0.037)	-0.272 (0.622)
5	PRAA, PRFT	0.108 (0.164)	0.078 (0.196)	0.025 (0.200)	0.010 (0.233)
6	OXLC	-0.236 (0.642)	-0.237 (0.844)	0.025 (0.304)	-0.236 (0.642)
7	FTNT	0.037 (0.068)	-0.065 (0.072)	0.017 (0.085)	-0.026 (0.102)
8	RFIL	4.564 (8.900)	0.047 (9.125)	2.424 (8.176)	4.564 (8.900)
9	FOX	-0.031 (0.058)	-0.021 (0.044)	-0.005 (0.018)	-0.010 (0.018)